Key Stage 3
MATHEMATICS

Mark Scheme

Term 1

KS 3

Baseline Test

Principal Examiners have prepared these mark schemes for these papers. These mark schemes have not, therefore, been through the normal process of standardising that would take place for live papers.

Further copies of this Mark Scheme are available from aqa.org.uk

Glossary for Mark Schemes

GCSE examinations are marked in such a way as to award positive achievement wherever possible. Thus, for GCSE Mathematics papers, marks are awarded under various categories.

If a student uses a method which is not explicitly covered by the mark scheme the same principles of marking should be applied. Credit should be given to any valid methods. Examiners should seek advice from their senior examiner if in any doubt.

M	Method marks are awarded for a correct method which could lead to a correct answer.
A	Accuracy marks are awarded when following on from a correct method. It is not necessary to always see the method. This can be implied.
B	Marks awarded independent of method.
ft	Follow through marks. Marks awarded for correct working following a mistake in an earlier step.
SC	Special case. Marks awarded within the scheme for a common misinterpretation which has some mathematical worth.
M dep	A method mark dependent on a previous method mark being awarded.
B dep	A mark that can only be awarded if a previous independent mark has been awarded.
oe	Or equivalent. Accept answers that are equivalent. eg accept 0.5 as well as $\frac{1}{2}$
[*a*, *b*]	Accept values between *a* and *b* inclusive.
3.14 …	Allow answers which begin 3.14 eg 3.14, 3.142, 3.1416
Use of brackets	It is not necessary to see the bracketed work to award the marks.

Examiners should consistently apply the following principles

Diagrams

Diagrams that have working on them should be treated like normal responses. If a diagram has been written on but the correct response is within the answer space, the work within the answer space should be marked. Working on diagrams that contradicts work within the answer space is not to be considered as choice but as working, and is not, therefore, penalised.

Responses which appear to come from incorrect methods

Whenever there is doubt as to whether a student has used an incorrect method to obtain an answer, as a general principle, the benefit of doubt must be given to the student. In cases where there is no doubt that the answer has come from incorrect working then the student should be penalised.

Questions which ask students to show working

Instructions on marking will be given but usually marks are not awarded to students who show no working.

Questions which do not ask students to show working

As a general principle, a correct response is awarded full marks.

Misread or miscopy

Students often copy values from a question incorrectly. If the examiner thinks that the student has made a genuine misread, then only the accuracy marks (A or B marks), up to a maximum of 2 marks are penalised. The method marks can still be awarded.

Further work

Once the correct answer has been seen, further working may be ignored unless it goes on to contradict the correct answer.

Choice

When a choice of answers and/or methods is given, mark each attempt. If both methods are valid then M marks can be awarded but any incorrect answer or method would result in marks being lost.

Work not replaced

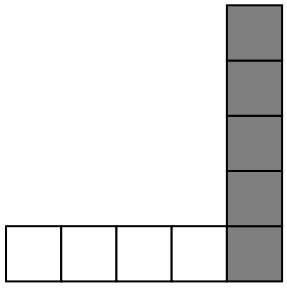
Erased or crossed out work that is still legible should be marked.

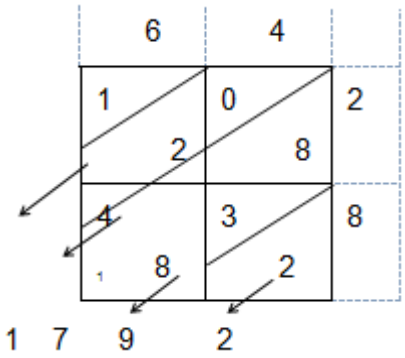
Work replaced

Erased or crossed out work that has been replaced is not awarded marks.

Premature approximation

Rounding off too early can lead to inaccuracy in the final answer. This should be penalised by 1 mark unless instructed otherwise.

Q	Answer	Mark	Comments
1	85	B1	
2(a)	2050 2500 20 005 25 000	B1	
2(b)	0.4 0.5 1 1.4	B1	
2(c)	$\frac{1}{2}$ $\frac{9}{10}$ 1 $\frac{5}{2}$	B1	
3(a)		B1	Any clear indication
3(b)	(White) 25	B1	
	(Grey) 26	B1ft	ft their 25 + 1
3(c)	Always an odd number added to an even number (which makes an odd number)	B1	oe eg1 Starts odd and adding two each time (which stays odd) eg2 Always one more grey than white so double and add one (which is always odd) eg3 $x + x + 1 = 2x + 1$ = odd
4(a)	50	B1	
4(b)	49	B2	B1 for any square number
5(a)	Maths **and** (History) 45 min **and** (Maths) 50 min	B2	B1 (History) 45 min **or** (Maths) 50 min
5(b)	(1.50 +) 1h 15 min	M1	$\frac{1}{4}$ h = 15 min
	3.05	A1	

Q	Answer	Mark	Comments
6(a)	126 (±2)	B1	
6(b)	360 – their 126	M1	
	234	A1ft	ft their 126 SC1 234 (±2) with no working seen
7	**Alternative method 1**		
	1280 **or** 512	M1	
	their 1280 + their 512	M1	Allow up to 2 errors in their numbers but must have 0 units in their 1280
	1792	A1	
	Alternative method 2		
	Four values with at least 3 correct from 1200 80 480 32	M1	
	their 1200 + their 80 + their 480 + their 32	M1	Allow up to 2 errors in their numbers
	1792	A1	
	Alternative method 3		
		M1 M1 A1	
8(a)	2589	B1	
8(b)	59.82	B1	
8(c)	(8 + 2) × (9 – 5)	B1	(2 + 8) × (9 – 5)

Q	Answer	Mark	Comments
9(a)	3200	B1	
9(b)	27	B1	
9(c)	4.8	B1	
9(d)	0.08	B1	
10	YES NO YES NO YES	B3	Allow one error, so B3 for 4 correct answers. B2 Any 3 correct B1 Any 2 correct
11	$75\% \rightarrow \frac{3}{4}$ $\frac{4}{6} \rightarrow \frac{2}{3}$ $\frac{3}{8} \rightarrow \frac{12}{32}$	B3	B2 2 correct B1 1 correct
12	All 5 bars with correct height **and** all bars same width **and** fully correct labelling on both axes **and** an equal gap between bars	B3	B2 Any 3 criteria satisfied B1 Any 2 criteria or any 1 criterion satisfied
13	30	B1	
14	3.7(0) -1.3(0)	M1	370 – 130 if correct
	2.4(0) ÷ 2	M1	240 ÷ 2
	£1.20	A1	120p £1.2 is M2A0

Q	Answer	Mark	Comments
15(a)	Attempt at sum ÷ 10	M1	120 ÷ 10 if correct
	1.2	A1	SC1 $1\frac{1}{5}$ or $\frac{6}{5}$ oe fraction
15(b)	0	B1	
16(a)	(–3, –2)	B1	
16(b)	(*k*, 1) marked on grid any *k* > –3	B1	
17(a)	4 by 3 **or** 6 by 1 rectangle drawn	B1	Do not accept a 5 by 2 rectangle in a different orientation
17(b)	4 by 4 square drawn	B1	
18	250 ÷ 5 (× 8) (= 50 (× 8))	M1	Any valid step eg 50 miles = 80 km
	400	A1	
19	< > = <	B4	B1 For each correct part
20	Only 3rd and 4th diagrams circled Second row	B2	B1 One or two circled with exactly one correct **or** Exactly three circled with the two correct ones included
21	4 tins seen or implied	B1	eg $2\frac{1}{2} + 2\frac{1}{2} + 2\frac{1}{2} + 2\frac{1}{2}$
	their 4 × 18	M1	their 4 must be > 1
	72	A1ft	ft from B0 M1 SC1 64.8(0)

Q	Answer	Mark	Comments
22	There are not the same number of each colour	B1	oe eg There are more blue discs